



Coevolution of opinion and consumption behavior under a two-layer network framework

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ABSTRACT

Individuals' optimistic or pessimistic perceptions of the economic outlook typically spread rapidly through online networks, while consumption intentions and behaviors play a critical feedback role in this process with active consumption boosting market confidence and cautious consumption reinforcing pessimistic expectations. To elucidate this complex coupling mechanism, we construct and analyze a dynamic model – within a two-layer network framework – that integrates opinion propagation with a consumption activation–deactivation process (i.e., transitioning between active and inactive consumption). Specifically, we employ a threshold model on the first layer to simulate the evolution of “positive/negative economic-outlook” opinions, and an Susceptible–Infected–Susceptible(SIS)-like model on the second layer to capture individuals' consumption behaviors. These two layers are then tightly coupled through an interaction formula. In experiments on a synthetically generated ER- k -regular two-layer network the system flips abruptly, jumping from a pessimistic-low-consumption state of roughly (0.16, 0.45) to an optimistic-high-consumption state of roughly (1.00, 0.95); for the real two-layer network derived from the Wiki-Vote dataset the deterministic and Monte-Carlo steady-state values differ by no more than 5 percent, and the phase-transition point shifts by only about 0.03, confirming a robust, quantifiable phase change. This sharp sensitivity defines a narrow policy-sensitive region where modest interventions can steer collective expectations and revive consumption.

1. Introduction

In recent years, with the rapid acceleration of information technology and globalization, economic activities and opinion dissemination in human society have increasingly exhibited networked and complex characteristics. Real-world economic systems, consumer behaviors, and opinion propagation processes can all be abstracted as complex networks, where nodes represent individuals or economic agents and edges depict the interactions and information flows between them. Complex network theory offers a novel perspective that not only reveals the intricate mechanisms underlying market sentiment, opinion diffusion, and consumption decisions, but also provides theoretical support for understanding and predicting economic fluctuations. By integrating economic activities and opinion dissemination within a unified framework, we can more precisely investigate the mutual influences

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of individual behaviors in collective decision-making and thus uncover the profound impact of network structure on the evolution of macroeconomic dynamics.

Despite significant progress made by single-layer network models in describing economic opinion diffusion and consumption dynamics, the complex interactions between economic opinions and consumption behavior within a multilayer network framework remain underexplored [1–3]. In particular, existing studies have not systematically investigated the coupling mechanisms in dual-layer networks of opinion and consumption, often overlooking how the interplay between threshold-based opinion propagation and consumption behavior jointly shapes the dynamic evolution of socioeconomic systems. This bidirectional coupling mechanism forms the dynamic interaction between opinion diffusion and consumption behavior, providing a novel theoretical framework that facilitates a deeper understanding of the co-evolution of opinions and consumption in economic contexts. Specifically, optimistic economic expectations are reinforced by active consumption behavior, which in turn lowers the threshold for opinion change, thereby promoting the widespread diffusion of positive views and creating a self-enhancing feedback loop. Understanding the coupling mechanism between opinion dissemination and consumption behavior is of critical importance for economic research.

Based on the above considerations, we propose a two-layer “Opinion–Consumption” coupled dynamical model by introducing a threshold mechanism tied to the baseline sentiment level. In this framework, the first-layer network employs a threshold model to characterize the individual’s shift in opinion about the economic outlook and to simulate the diffusion of economic opinions among agents; the second-layer network, modeled in a manner analogous to a Susceptible–Infected–Susceptible (SIS) process, describes the activation and withdrawal of consumption behavior, thereby revealing how consumption willingness fluctuates in response to opinion changes [4–6]. By coupling these two network layers, we can thoroughly investigate the mutual feedback mechanism between opinion diffusion and consumption behavior. Furthermore, we investigate the effect of key parameters in three scenarios: a synthetic network with an Erdős–Rényi(ER) opinion layer and a k -regular consumption layer (ER– k -regular network), a synthetic network in which both layers are k -regular(k -regular two-layer network), and an empirical case in which the opinion and consumption layers are mapped directly onto the *Wiki-Vote* network(*Wiki-Vote* two-layer network). Numerical results indicate that when positive feedback is established between consumption intensity and the opinion threshold, the system can undergo a rapid transition from a “pessimistic-low consumption” state to an “optimistic-high consumption” state, exhibiting a pronounced phase transition. This finding not only offers a new perspective on the nonlinear coupled evolution between macro-economic confidence and consumption behavior, but also provides potential theoretical support for policy interventions and economic regulation. There are three main contributions to this study:

(1) Online opinion diffusion and offline consumption are coupled in a two-layer network that combines threshold-based adoption with SIS-style activation, enabling integrated analysis of information and behavioral flows.

(2) Across both synthetic architectures (ER– k -regular network and k -regular two-layer network) and the empirical setting (*Wiki-Vote* two-layer network), the experiments show that varying the baseline sentiment level induces a pronounced phase transition.

(3) Incorporates economic outlook and consumption for policy or stimulus guidance. The model captures how optimistic (or pessimistic) expectations reinforce (or weaken) consumption intensity, offering a quantitative tool for assessing macro-policy interventions.

The rest of the paper is organized as follows: Section 3 introduces the methods for network construction and formulates the coupled two-layer network model together with the mathematical expressions governing opinion diffusion and behavioral propagation. Section 4 reports simulation experiments on the ER– k -regular network and k -regular two-layer network, as well as on the *Wiki-Vote* two-layer network, and provides corresponding visualizations and discussions. Finally, Sections 5 and 6 summarize the discussion and conclusions.

2. Related work

Opinion diffusion has long been a central topic in the study of social dynamics and information propagation [7,8]. Research on opinion diffusion originates from multiple disciplines – including sociology, physics, epidemiology, and economics – and spans diverse phenomena ranging from the spread of social trust to the transmission of emotions, beliefs, and economic opinions [9–11]. In networked systems, the process of opinion diffusion typically involves the transformation of individual attitudes from negative to positive. This process illustrates the fundamental mechanism of mutual influence among agents in social networks. It also helps clarify how economic entities are affected by the behaviors and opinions of others, which can lead to rapid changes in market sentiment, consumption preferences, and economic decisions [12–15]. However, most existing studies treat opinion diffusion as an independent process, leaving its interplay with subsequent consumption behavior largely under-explored.

Within the propagation process, the consensus and divergence emerging from opinion diffusion are critical research areas because they reflect the uniformity of social cognition and, in turn, influence the decisions of economic agents and market trends [16]. Consensus refers to a situation where the majority of network members converge on a common opinion, often manifesting as a convergence of consumer preferences and concentrated market demand. In contrast, divergence is characterized by opposing views among individuals, potentially leading to market segmentation, differentiated consumption, or even imbalanced resource allocation. Numerous studies have demonstrated that reaching consensus is a complex and challenging process, especially in economic environments marked by diverse consumer preferences and highly heterogeneous market demands [17–20]. Differences in economic background, beliefs, and cultural factors not only lead to a variety of decision-making preferences, but also hinder the convergence of collective decisions and consumer opinions, thereby intensifying opinion divergence [21–23]. From an economic

standpoint, both consensus and divergence affect not only the uniformity of social opinions but also individual economic decision-making. In particular, when it comes to consumption behavior, choices regarding consumption trends, brand preferences, and purchasing decisions are often driven by prevailing opinions or collective consensus (or divergence) within the network. Yet, how such consensus or divergence quantitatively feeds back into consumption dynamics across coupled network layers remains unclear.

The study of behavior in complex networks has emerged as a hot topic that spans multiple disciplines, covering a wide range of individual to collective behaviors [24,25]. In social networks, individual behavior is influenced not only by personal motives and preferences but also, to a large extent, by the behavior of others within the network. From an economic perspective, such interactions often manifest as convergence or divergence in consumption or investment preferences. Behavioral propagation is typically viewed as a diffusion process of information or states, where an individual's state change is influenced by the behavior of their neighbors or social circles [26–28]. When the majority of individuals behave similarly, their collective action exerts a strong influence that drives the entire system in a specific direction [29,30]. For instance, in the contexts of consumption, voting, or the spread of popular culture, individual choices are often significantly influenced by those around them, leading to what is known as social behavior diffusion [31–33]. Moreover, the impact of the macroeconomic outlook on individual consumption behavior has become a core issue in economic research [34–36]. In economic research, the diffusion of consumption behavior has long been a central issue, particularly in understanding how individual purchasing decisions propagate through social networks and shape market trends. Studying this diffusion is essential for explaining collective consumption waves, predicting changes in aggregate demand, and formulating policies that can mitigate adverse cascade effects while promoting beneficial adoption [37,38]. However, relatively few studies have integrated this topic with complex network theory. The complex network framework offers a powerful tool for modeling these processes, capturing peer effects and social influence typical in consumption dynamics. For example, Ref. [39] modeled the diffusion of new product consumption within a complex network framework and empirically validated the model. Ref. [40] incorporated a recovery mechanism into consumption diffusion under the same framework, proposing a hybrid Bass-SIR model to simulate the spread of purchasing behavior and verifying its validity through simulations. Similarly, Ref. [41] applied complex network methods to model the adoption and usage of new pharmaceuticals, testing contagion effects against empirical data. This integration allows a more comprehensive analysis of how micro-level behaviors aggregate into macro phenomena and underscores the value of network-based approaches in economic modeling. That said, a unified framework that combines behavioral diffusion with opinion-driven economic expectations and captures their bidirectional reinforcement remains underdeveloped.

3. Models and methods

3.1. Construction of a two-layer network structure

In this study, we develop a two-layer network model to investigate the co-evolutionary mechanisms between opinion and consumption behavior within an economic environment. The model comprises two networks with independent topologies yet featuring a one-to-one correspondence between their nodes, representing distinct social relationships: the online Opinion Layer and the offline Consumption Layer. This two-layer structure not only captures the dissemination phenomena occurring within the same group through different modes of social interaction, but also unveils the complex feedback mechanisms between opinions and consumption behaviors.

As shown in Fig. 1, in the Opinion Layer, each node represents an economic agent whose state reflects its optimistic or pessimistic outlook on the economy. The edges between nodes denote interactions such as information exchange, opinion influence, and emotional contagion, thereby simulating the spread of “positive” or “negative” opinions throughout the network. By tracking the temporal evolution of node states, one can quantify the dynamic distribution of public sentiment and assess its potential impact on subsequent economic decision-making.

The nodes in the Consumption Layer correspond one-to-one with those in the Opinion Layer. In this layer, each node's state indicates the level of consumer activity (e.g. active consumer or inactive consumer), while the edges capture interactions related to consumption tendencies and market feedback. When an individual shifts from inactivity to active consumption, it often exerts a demonstrative or catalytic effect on its neighbors; conversely, when an individual withdraws or reduces its consumption propensity, it may amplify cautious sentiment among surrounding agents.

The independent topological design of the two layers ensures that opinion diffusion and consumption behavior evolve separately within their respective social structures, while the dynamic mutual feedback between the Opinion and Consumption Layers enables their dynamic coupling.

3.2. Opinion diffusion

Let $\mu_{p,t}$ denote the fraction of nodes in the opinion layer holding a positive opinion at time t . We can determine the probability R_t that a randomly chosen agent with an active opinion can be reached by another randomly chosen agent with an inactive opinion. We employ a threshold model in which individuals adjust their opinions under the combined influence of global information and local cues. When the integrated signal received by an individual exceeds a certain threshold, its opinion flips. This modeling choice is preferred over continuous formulations (e.g., logistic or linear response functions) because it effectively captures abrupt shifts in collective sentiment, which are commonly observed in real-world opinion dynamics, such as in the spread of innovations, financial bubbles, or social movements. Threshold models are particularly suited for scenarios where individuals need to reach a critical point of influence before changing their behavior, leading to cascading effects that are often smoothed over by continuous models. For

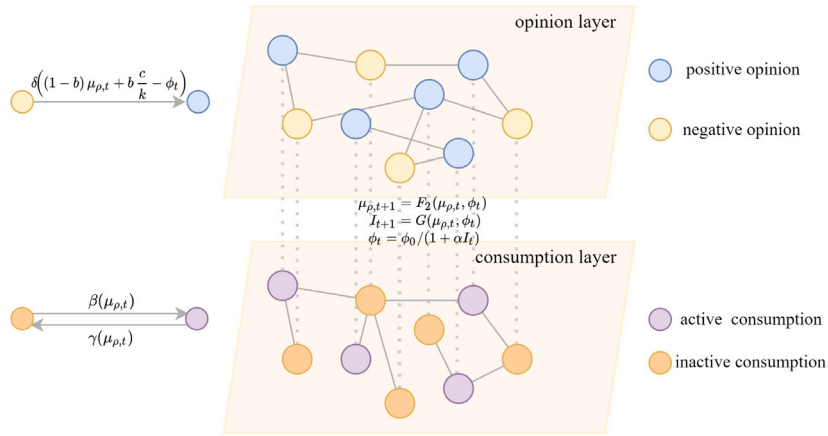


Fig. 1. Schematic of the two-layer network. (Blue circles denote nodes with an positive opinion, yellow circles denote negative opinion, purple circles denote active consumers, and orange circles denote inactive consumers. On the opinion layer, a yellow node switches to blue according to the indicator function $\delta((1-b)\mu_{\rho,t} + b\frac{c}{k} - \phi_t)$, which equals 1 when the integrated signal exceeds the threshold ϕ_t . On the consumption layer, an orange node becomes purple with probability $\beta(\mu_{\rho,t})$, and a purple node reverts to orange with probability $\gamma(\mu_{\rho,t})$. The two layers are coupled via $\mu_{\rho,t+1} = F_2(\mu_{\rho,t}, \phi_t)$ and $I_{t+1} = G(\mu_{\rho,t}, \phi_t)$.

example, in [42] a large-scale field experiment on online health behavior shows that adoption often requires multiple simultaneous confirmations, and this phenomenon can be well captured by threshold rules, but linear or logistic response models cannot reflect this pattern. Empirical studies in social dynamics and network science have demonstrated the validity of threshold-based approaches for modeling opinion cascades and phase transitions [43,44]. Specifically, for a given node, if the weighted combination of the global optimism $\mu_{\rho,t}$ and the fraction of its neighbors holding positive opinions exceeds the threshold ϕ_t , then that node switches to a positive opinion. Building on formulations from both microscopic and macroscopic dynamics [45–48], we propose:

$$R_{t+1} = r + (1-r) \sum_{k=1}^{\infty} \frac{k p(k)}{\langle k \rangle} \sum_{c=0}^{k-1} \binom{k-1}{c} R_t^c (1-R_t)^{k-1-c} \delta((1-b)\mu_{\rho,t} + b\frac{c}{k} - \phi_t) \triangleq F_1(\mu_{\rho,t}, \phi_t), \quad (1)$$

$$\mu_{\rho,t+1} = r + (1-r) \sum_{k=0}^{\infty} p(k) \sum_{c=0}^k \binom{k}{c} R_t^c (1-R_t)^{k-c} \delta((1-b)\mu_{\rho,t} + b\frac{c}{k} - \phi_t) \triangleq F_2(\mu_{\rho,t}, \phi_t). \quad (2)$$

where r is the initial probability that a node spontaneously adopts a positive opinion. $F_1(\mu_{\rho,t}, I_t)$ denotes that R_{t+1} is defined as a function of $\mu_{\rho,t}$ and I_t , and similarly for $F_2(\mu_{\rho,t}, I_t)$. The term $p(k)$ represents the degree distribution of the opinion layer, and $\langle k \rangle$ is its average degree; the factor $\frac{k p(k)}{\langle k \rangle}$ thus denotes the probability that a randomly chosen edge is connected to a node with degree k . For a node of degree k , the inner sums account for the probability that exactly c out of its $(k-1$ or k , respectively) neighbors are positive—given by the binomial terms $\binom{k-1}{c} R_t^c (1-R_t)^{k-1-c}$ and $\binom{k}{c} R_t^c (1-R_t)^{k-c}$. Here, c consistently represents “the number of neighbors holding a positive opinion”, but the differing upper bounds arise from distinct sampling perspectives in the two update rules. In the update of R_{t+1} , we arrive at a node of degree k along a randomly selected edge. This incoming edge is fixed by construction, so only the remaining $k-1$ neighbors may vary. Thus, $c \in \{0, \dots, k-1\}$. In the update of $\mu_{\rho,t+1}$, we uniformly sample a node at random. In this case, all k of its neighbors contribute, so $c \in \{0, \dots, k\}$. The indicator function $\delta((1-b)\mu_{\rho,t} + b\frac{c}{k} - \phi_t)$ equals 1 if the integrated signal, a weighted sum of the global opinion $\mu_{\rho,t}$ and the local fraction $\frac{c}{k}$, exceeds the threshold ϕ_t , and 0 otherwise. The parameter b (with $0 \leq b \leq 1$) governs the relative influence of global versus local information.

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3.3. Consumption behavior

Let I_t denote the fraction of active consumers in the consumption layer at time t . We model the evolution of this layer using a discrete SIS framework, which captures the transitions between participation (active consumption) and withdrawal (inactive consumption). $G(\mu_{\rho,t}, \phi_t)$ denotes that I_{t+1} is defined as a function of $\mu_{\rho,t}$ and I_t . The dynamic evolution is given by

$$I_{t+1} = I_t + \beta(\mu_{\rho,t}) \langle k_2 \rangle I_t [1 - I_t] - \gamma(\mu_{\rho,t}) I_t \triangleq G(\mu_{\rho,t}, \phi_t), \quad (3)$$

where the activation rate $\beta(\mu_{\rho,t})$ and the exit rate $\gamma(\mu_{\rho,t})$ are functions of the current opinion-layer indicator $\mu_{\rho,t}$ and, based on the considerations in [49], to ensure that I_t evolves within $[0, 1]$, satisfy the condition $\beta(\mu_{\rho,t}) \langle k_2 \rangle \leq 1 + \gamma(\mu_{\rho,t})$. In particular, these functions

are defined as

$$\beta(\mu_{p,t}) = \beta_0 + \beta_1 \mu_{p,t}, \quad \gamma(\mu_{p,t}) = \gamma_0 - \gamma_1 \mu_{p,t}. \quad (4)$$

where $\beta(\mu_{p,t})$ represents the rate at which inactive consumers become active, reflecting the sensitivity of consumer behavior to economic optimism. Conversely, $\gamma(\mu_{p,t})$ denotes the rate at which active consumers exit, capturing the tendency for individuals to withdraw from active consumption when economic conditions worsen. The parameter $\langle k_2 \rangle$ is the average degree of the consumption network.

Together, these equations encapsulate the interplay between public sentiment and consumer participation, illustrating how shifts in economic outlook mediated by the opinion layer can influence the transition between active and inactive consumption states.

3.4. The two-way coupling of opinion and consumption

Two-layer network coupling is achieved through ϕ and I , with the coupling core centered on the threshold:

$$\phi_t = \frac{\phi_0}{1 + \alpha I_t}, \quad (5)$$

Here, ϕ_0 denotes the initial threshold, representing the baseline sentiment level, for opinion change in the absence of consumption behavior effects. The parameter α is the modulation coefficient by which consumption activity adjusts the threshold in the opinion layer, thereby determining the strength of consumption behavior's impact on opinion propagation. As α increases, ϕ_t decreases, which implies that high consumption activity lowers the barrier for individuals in the opinion layer to adopt a “positive” stance, thus facilitating the spread of optimism. This mechanism exemplifies the positive feedback effect of economic activity on social expectations, significantly enhancing the inter-layer interactions within certain parameter ranges and leading to nonlinear changes in system behavior. In other words, the higher the I , the lower the threshold becomes, making individuals more prone to accept a positive opinion and rendering the opinion layer more susceptible to cascades of positive views—a “positive feedback” loop that often reinforces the interplay between opinions and consumption under suitable parameter settings.

The core of this model lies in its bidirectional coupling mechanism, which captures the dynamic mutual feedback between the opinion and consumption layers. Specifically, the consumption layer influences the opinion layer by dynamically reducing the threshold ϕ_t through consumption activity I_t , thereby facilitating the spread of positive opinions. This mechanism reflects how active consumption behaviors reinforce individuals' optimistic expectations regarding economic conditions, ultimately enhancing overall economic confidence. Conversely, the opinion layer impacts the consumption layer through the proportion of positive opinions $\mu_{p,t}$, which modulates the activation rate $\beta(\mu_{p,t})$ in the consumption layer. In other words, increased economic optimism strengthens incentives for consumption, highlighting the reinforcing effect of economic confidence on consumption dynamics.

3.5. Real-world correspondence

In [48], to ensure that the state variables of the model evolve within the interval $[0, 1]$, all probabilities, fractions, and thresholds are normalized to the unit interval $[0, 1]$. In the present study, the baseline sentiment level $\phi_0 \in [0, 1]$ represents the threshold for adopting a positive opinion in the absence of consumption feedback. It enters the time-varying threshold via $\phi_t = \phi_0/(1 + \alpha I_t)$. Setting $\phi_0 = 0$ corresponds to nearly barrier-free opinion switching, where almost every node changes its opinion, while $\phi_0 = 1$ represents the most resistant scenario, where almost no node changes its opinion. A larger ϕ_0 implies greater difficulty in opinion change. The weight $b \in [0, 1]$ balances global and local influences in the opinion update rule: $b = 0$ means the response is entirely driven by the global optimism level $\mu_{p,t}$, $b = 1$ implies complete reliance on the local fraction of positive neighbors c/k , and intermediate values interpolate between these two sources of influence. The parameter $r \in [0, 1]$ quantifies the initial proportion of optimistic nodes. $r = 0$ indicates all nodes initially hold a negative opinion, while $r = 1$ means all nodes begin with a positive opinion. The feedback coefficient $\alpha \geq 0$ controls the strength with which consumption activity lowers the opinion threshold via $\phi_t = \phi_0/(1 + \alpha I_t)$. When $\alpha = 0$, the consumption layer has minimal influence on the opinion layer; larger values of α correspond to stronger influence.

In the consumption layer, the activation rate of inactive consumers is given by $\beta(\mu_{p,t}) = \beta_0 + \beta_1 \mu_{p,t}$, and the exit rate of active consumers is given by $\gamma(\mu_{p,t}) = \gamma_0 - \gamma_1 \mu_{p,t}$. Here, β_0 represents the baseline activation rate in the absence of opinion influence, and γ_0 denotes the baseline exit rate without opinion effects. The coefficients β_1 and γ_1 capture the weight of opinion influence on consumption behavior. We take $\beta_0, \gamma_0 \in [0, 1]$ and choose β_1, γ_1 such that $\beta(\mu_{p,t}), \gamma(\mu_{p,t}) \in [0, 1]$ for all $\mu_{p,t} \in [0, 1]$. Additionally, we enforce the discrete-time constraint $\beta(\mu_{p,t})\langle k_2 \rangle \leq 1 + \gamma(\mu_{p,t})$ for all $\mu_{p,t} \in [0, 1]$.

3.6. Fixed points and self-consistent equations

We can prove that the system admits fixed points, see [Appendix](#) for details.

3.6.1. Fixed-point condition

The fixed-point update equation for I_t is

$$I_{t+1} = I_t + \beta(\mu_\rho)\langle k^2 \rangle I_t(1 - I_t) - \gamma(\mu_\rho) I_t$$

The fixed-point condition is

$$I_{t+1} = I_t$$

Substituting gives

$$\beta(\mu_\rho)\langle k^2 \rangle I(1 - I) - \gamma(\mu_\rho) I = 0$$

Factoring out I :

$$I [\beta(\mu_\rho)\langle k^2 \rangle (1 - I) - \gamma(\mu_\rho)] = 0$$

We discuss two cases:

1. Trivial solution: $I = 0$.
2. Non-trivial solution requires

$$\beta(\mu_\rho)\langle k^2 \rangle (1 - I) = \gamma(\mu_\rho),$$

$$\beta(\mu_\rho)\langle k^2 \rangle - \beta(\mu_\rho)\langle k^2 \rangle I = \gamma(\mu_\rho),$$

$$I = \frac{\beta(\mu_\rho)\langle k^2 \rangle - \gamma(\mu_\rho)}{\beta(\mu_\rho)\langle k^2 \rangle} = 1 - \frac{\gamma(\mu_\rho)}{\beta(\mu_\rho)\langle k^2 \rangle}.$$

The fixed-point condition indicates that the system reaches a steady state where the proportion of active consumers does not change over time. The trivial solution $I = 0$ corresponds to an economic equilibrium where all consumers are inactive, indicating no consumption activity (economic stagnation). The non-trivial solution represents an endemic state where a constant proportion of consumers remain active, reflecting sustained economic activity. The expression for I shows that the steady-state level depends on the parameters β and γ , which are influenced by the mean opinion μ_ρ from the Opinion Layer, highlighting the coupling between opinion diffusion and consumption behavior. Substituting the definitions

$$\beta(\mu_\rho) = \beta_0 + \beta_1 \mu_\rho, \quad \gamma(\mu_\rho) = \gamma_0 - \gamma_1 \mu_\rho,$$

we finally obtain the non-trivial solution

$$I^* = 1 - \frac{\gamma_0 - \gamma_1 \mu_\rho}{(\beta_0 + \beta_1 \mu_\rho)\langle k^2 \rangle} = 1 - \frac{\gamma_0 - \gamma_1 \mu_\rho^*}{(\beta_0 + \beta_1 \mu_\rho^*)\langle k^2 \rangle}.$$

This solution I^* explicitly shows how the steady-state proportion of active consumers is affected by the mean opinion μ_ρ^* . Here, $\beta(\mu_\rho)$ represents the consumption activation rate, which increases with more optimistic opinions (higher μ_ρ), while $\gamma(\mu_\rho)$ represents the consumption deactivation rate, which decreases with more optimistic opinions. Thus, as μ_ρ^* increases, the effective activation rate increases and the deactivation rate decreases, leading to a higher steady-state level of active consumers if other parameters are fixed. This underscores the role of public opinion in modulating consumption dynamics, where optimism stimulates economic activity.

3.6.2. Expression for ϕ^* , μ_ρ^* and R^*

Since ϕ_t appears in the updates of μ_ρ and R , and

$$\phi_t = \frac{\phi_0}{1 + \alpha I_t},$$

at the fixed point we have

$$\phi = \frac{\phi_0}{1 + \alpha I^*}.$$

Substituting the above I^* expresses ϕ as a function of μ_ρ :

$$\phi^* = \frac{\phi_0}{1 + \alpha I^*}.$$

Thus a higher activity level lowers the effective threshold ϕ^* when $\alpha > 0$, creating a positive feedback: larger I^* makes adoption easier, which in turn sustains larger μ_ρ^* and R^* .

$$\mu_\rho^* = r + (1 - r) \sum_{k=0}^{\infty} p(k) \sum_{c=0}^k \binom{k}{c} (R^*)^c (1 - R^*)^{k-c} \delta \left((1 - b) \mu_\rho^* + b \frac{c}{k} - \phi^* \right).$$

Note that here $\delta(\cdot)$ is the Heaviside step function:

$$\delta(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Only the terms satisfying

$$(1-b)\mu_\rho^* + b\frac{c}{k} \geq \phi^*$$

contribute.

$$R^* = r + (1-r) \sum_{k=1}^{\infty} \frac{k p(k)}{\langle k_1 \rangle} \sum_{c=0}^{k-1} \binom{k-1}{c} (R^*)^c (1-R^*)^{k-1-c} \delta\left((1-b)\mu_\rho^* + b\frac{c}{k} - \phi^*\right).$$

Only terms with

$$(1-b)\mu_\rho^* + b\frac{c}{k} \geq \phi^*$$

contribute.

Therefore, the complete system is

$$I^* = 1 - \frac{\gamma_0 - \gamma_1 \mu_\rho^*}{(\beta_0 + \beta_1 \mu_\rho^*) \langle k^2 \rangle},$$

$$\phi^* = \frac{\phi_0}{1 + \alpha I^*},$$

$$\mu_\rho^* = r + (1-r) \sum_{k=0}^{\infty} p(k) \sum_{c=0}^k \binom{k}{c} (R^*)^c (1-R^*)^{k-c} \delta\left((1-b)\mu_\rho^* + b\frac{c}{k} - \phi^*\right),$$

$$R^* = r + (1-r) \sum_{k=1}^{\infty} \frac{k p(k)}{\langle k_1 \rangle} \sum_{c=0}^{k-1} \binom{k-1}{c} (R^*)^c (1-R^*)^{k-1-c} \delta\left((1-b)\mu_\rho^* + b\frac{c}{k} - \phi^*\right).$$

These three coupled equations make the feedback explicit: I^* influences ϕ^* , which shifts the effective threshold inside the μ_ρ^* and R^* equations; the resulting μ_ρ^* in turn feeds back into $\beta(\cdot)$ and $\gamma(\cdot)$ that determine I^* . Depending on parameters, the system can admit a unique fixed point or multiple admissible solutions; in practice, fixed-point iteration converges rapidly in the parameter ranges we study.

3.6.3. Threshold and cumulative distribution functions

$$\phi^* = \frac{\phi_0}{1 + \alpha I^*}, \quad T = \frac{\phi^* - (1-b)\mu_\rho^*}{b}.$$

Let

$$\Psi(R^*; T) = \sum_{k=0}^{\infty} p(k) \sum_{c=\lceil kT \rceil}^k \binom{k}{c} (R^*)^c (1-R^*)^{k-c},$$

$$\Phi(R^*; T) = \sum_{k=1}^{\infty} \frac{k p(k)}{\langle k_1 \rangle} \sum_{c=\lceil kT \rceil}^{k-1} \binom{k-1}{c} (R^*)^c (1-R^*)^{k-1-c}.$$

The threshold T is derived from ϕ^* and μ_ρ^* , and it represents the critical value above which nodes in the Opinion Layer adopt the opinion (e.g., become optimistic). The functions Ψ and Φ are cumulative distribution functions that aggregate the probability over the degree distribution and binomial outcomes, capturing the proportion of nodes that meet the threshold condition. This formulation allows us to compute the steady-state values numerically or analytically for specific distributions, providing a bridge between individual behavior and population-level dynamics in both opinion and consumption layers.

3.6.4. Self-consistent equations

$$I^* = 0 \quad \text{or} \quad I^* = 1 - \frac{\gamma_0 - \gamma_1 \mu_\rho^*}{(\beta_0 + \beta_1 \mu_\rho^*) \langle k^2 \rangle},$$

$$\mu_\rho^* = r + (1-r) \Psi(R^*; T(I^*, \mu_\rho^*)),$$

$$R^* = r + (1-r) \Phi(R^*; T(I^*, \mu_\rho^*)),$$

where

$$T(I^*, \mu_\rho^*) = \frac{\frac{\phi_0}{1 + \alpha I^*} - (1-b)\mu_\rho^*}{b}.$$

Table 1

Model parameters, admissible ranges, and benchmark settings under two network configurations. ER- k -regular: opinion layer is Erdős-Rényi with $\langle k_1 \rangle = 4$, consumption layer is k -regular with $k = \langle k_2 \rangle = 4$. k -regular-two-layer network: both layers are k -regular with $k = 4$. All probabilities, fractions, and thresholds are normalized to $[0, 1]$.

Symbol	Name	Range/Constraint	ER- k -reg	k - k -reg	Notes
N	network size	$\mathbb{N}_+$	1000	1000	balances finite-size and runtime
$\langle k_1 \rangle$	mean degree (opinion)	> 0	4	4	from network construction
$\langle k_2 \rangle$	mean degree (consumption)	> 0	4	4	k -regular layer
$p(k)$	degree distribution (opinion)	$p(k) \in [0, 1], \sum_k p(k) = 1$	ER	k -regular	PMF fixed by construction
ϕ_0	baseline sentiment level	$[0, 1]$	sweep $[0, 1]$	sweep $[0, 1]$	$\phi_t = \phi_0 / (1 + \alpha I_t)$
α	threshold feedback strength	$\alpha \geq 0$ (practical $[0, 1]$)	sweep $[0, 1]$	sweep $[0, 1]$	larger α lowers ϕ_t , faster
b	global-local weight (opinion rule)	$[0, 1]$	sweep $[0, 1]$	sweep $[0, 1]$	$0 : \mu_{p,i}$ only; $1 : c/k$ only
r	seed fraction (spontaneous adoption)	$[0, 1]$	0.1	0.1	small initial optimism
$\beta(\mu)$	activation rate	$\beta_0 + \beta_1 \mu$	–	–	inactive→active
$\gamma(\mu)$	exit rate	$\gamma_0 - \gamma_1 \mu$	–	–	active→inactive
β_0	baseline activation	$[0, 1]$	0.07	0.07	kept within $[0, 1]$
β_1	optimism elasticity (activation)	$[0, 1]$	0.05	0.05	ensures $\beta(\mu) \in [0, 1]$
γ_0	baseline exit	$[0, 1]$	0.15	0.15	kept within $[0, 1]$
γ_1	optimism elasticity (retention)	$[0, 1]$	0.10	0.10	ensures $\gamma(\mu) \in [0, 1]$

Condition: $\beta(\mu) \langle k_2 \rangle \leq 1 + \gamma(\mu)$ for all $\mu \in [0, 1]$.

Taken together, these relations summarize the fixed point as a balance between (i) activity sustained by opinion-driven transmission and network heterogeneity, and (ii) recovery and thresholds that gate local adoption. This framework provides a comprehensive description of the coupled dynamics, where opinion spread influences consumption behavior and vice versa, leading to rich economic phases such as booms or recessions.

3.6.5. Poisson case

If $p(k) = e^{-\lambda} \lambda^k / k!$, then

$$\Psi(R^*; T) = \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} [1 - F_{\text{Bin}(k, R^*)}(\lceil kT \rceil - 1)],$$

where

$$F_{\text{Bin}(k, p)}(m) = \sum_{c=0}^m \binom{k}{c} p^c (1-p)^{k-c}.$$

Similarly,

$$\Phi(R^*; T) = \sum_{k=1}^{\infty} \frac{k e^{-\lambda} \lambda^k}{\langle k_1 \rangle k!} [1 - F_{\text{Bin}(k-1, R^*)}(\lceil kT \rceil - 1)].$$

Assuming a Poisson degree distribution gives us a simple, standard baseline. It keeps the math manageable and the computations fast, so fixed-point checks and simulations are easy to run. Real social networks are often more uneven, but this homogeneous-degree approximation still captures the core, mean-field behavior. It lets us see clearly how network density and local thresholds drive the joint evolution of opinions and behavior—without adding unnecessary complexity.

4. Experiments

4.1. Experimental setup and parameter description

4.1.1. Experimental setup

In the ER- k -regular network, to incorporate randomness, we adopt an Erdős-Rényi network as the opinion layer. Simultaneously, to simplify the model and ensure network uniformity, we employ a k -regular network as the consumption layer. Subsequently, we completely eliminate topological heterogeneity in the two-layer- k -regular network, meaning both layers are k -regular. The parameters of the two synthetic networks are kept consistent, allowing for direct comparisons to examine the effect of topological heterogeneity on the results. The third set of experiments is conducted on a real network structure, using a Wiki-Vote double-layer network. We map both layers onto the same underlying real network to preserve vertex correspondence, thereby investigating whether similar phenomena persist in real-world network structures. In this experiment, the dataset provides only topological information and contains no data on opinions or consumption. To avoid bias introduced by initial conditions or numerical methods, we report both deterministic iteration results and Monte Carlo simulation outcomes. Overall, experiments on the two synthetic networks provide preliminary validation of the model, and further confirm its effectiveness in *real* networks.

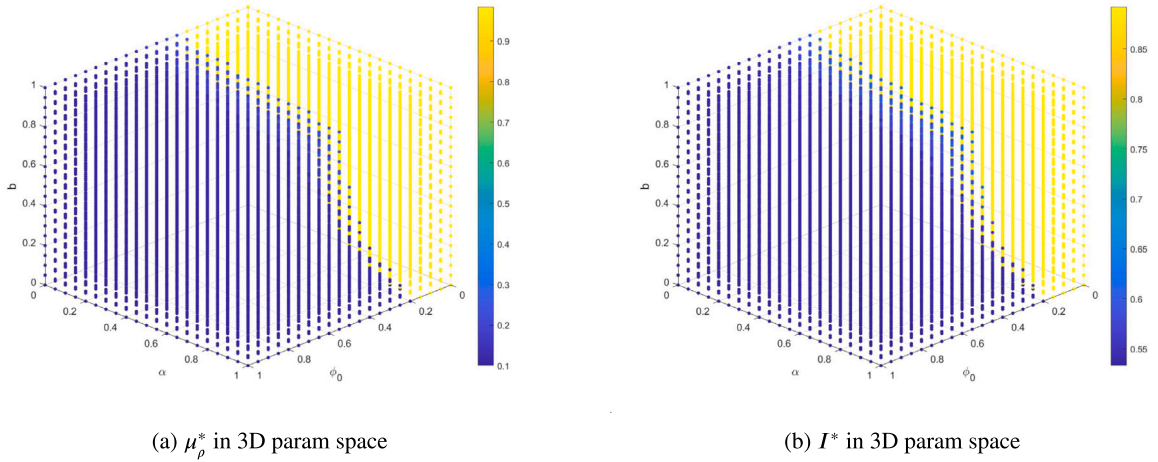


Fig. 2. Schematic of steady-state distributions (The X -axis represents the parameter α , the Y -axis represents the parameter ϕ_0 , the Z -axis represents the parameter b , and the color indicates the final steady-state proportion of good opinions μ_ρ^* or the fraction of active consumers I^* .)

4.1.2. Parameter description

Each layer consists of $N = 1000$ nodes, a size chosen to ensure result stability without excessive computational cost. The consumption layer is constructed as a k -regular network ($k = 4$), where every node has the same number of neighbors. The opinion layer is modeled as an Erdős-Rényi (ER) network with an average degree of $\langle k_1 \rangle = 4$. Initially, 10% of the nodes are randomly selected as optimists. Opinion dynamics evolve via a threshold rule that incorporates baseline sentiment ϕ_0 , feedback coefficient α , and local weight b . Consumption behavior is updated using an SIS-like rule. To prevent overly rapid convergence and to satisfy the condition $\beta(\mu)\langle k_2 \rangle \leq 1 + \gamma(\mu)$ for dynamics confined to $[0, 1]$, we set $\beta_0 = 0.07$, $\gamma_0 = 0.15$, $\beta_1 = 0.05$, and $\gamma_1 = 0.10$. These values reflect a conservative consumption tendency during economic downturns, where opinions exert relatively minor influence on consumption behavior. In the experiments, relevant parameters are varied over $[0, 1]$ in steps of 0.05. Simulations terminate when the change between two consecutive iterations falls below 10^{-6} , indicating convergence to a steady state. The same procedure is repeated in the second set of experiments, except that the ER opinion layer is replaced with another k -regular layer ($k = 4$), thereby completely removing topological differences. Parameter ranges and configurations are summarized in Table 1.

4.2. Model with the opinion layer as an Erdős-Rényi network

Fig. 2 displays the steady-state distributions obtained by performing a grid scan over the three-dimensional parameter space (ϕ_0, α, b) . It is evident that the proportion of “positive opinions”, μ_ρ^* , and the consumption activity level, I^* , exhibit pronounced polarized regions under different parameter combinations. The color scale, ranging from dark blue (lower values) to bright yellow (higher values), corresponds to a possible transition of the system from a “pessimistic-low consumption” state to an “optimistic-active consumption” state, which is a characteristic feature of the phase transition between threshold-driven opinion diffusion and consumption activation-withdrawal dynamics.

Specifically, when ϕ_0 is relatively small (indicating a lower initial opinion threshold) and α is large (meaning that increased consumption significantly reduces the opinion threshold), the system is more prone to cross the critical condition, driving the vast majority of agents toward positive opinions and triggering a rapid rise in consumption (as indicated by the bright yellow regions in the figures). In contrast, if ϕ_0 is high or α is low, the conditions required for opinion switching become more demanding, and the system typically remains in a local steady state characterized by a low proportion of positive opinions and low consumption (the blue regions in the figures). The parameter b regulates the influence of local neighbors on individual decision-making. When b is large, an initially optimistic cluster can rapidly propagate its cascade effect throughout the network. In contrast, when b is small, the global average predominates in individual decisions, necessitating more significant overall changes to drive a large-scale transition.

Both the μ_ρ^* and I^* figures exhibit a distinct blue–yellow partition, indicating a strong positive feedback coupling between opinions and consumption—optimism boosts consumption, and high consumption. Specifically, optimism stimulates consumption, while increased consumption subsequently lowers the opinion threshold, encouraging more agents to adopt an optimistic outlook. This positive feedback may trigger a phase transition over a wide range of parameters. It offers not only a mechanistic explanation for phenomena such as abrupt shifts in economic confidence or reversals in collective expectations, but it also provides crucial numerical evidence for policy interventions or strategies aimed at influencing opinions, which could help break the pessimistic low-consumption trap and stimulate a high-activity state.

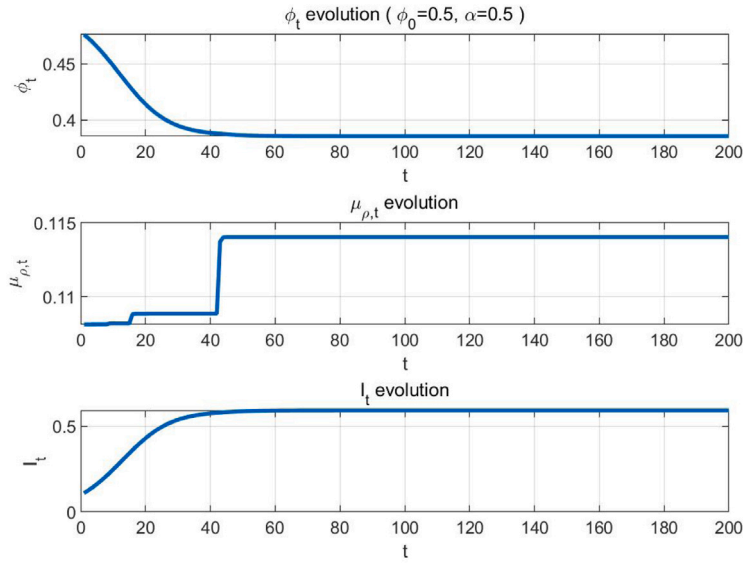


Fig. 3. Time evolution of the two-layer network : ϕ_t , $\mu_{\rho,t}$, I_t .

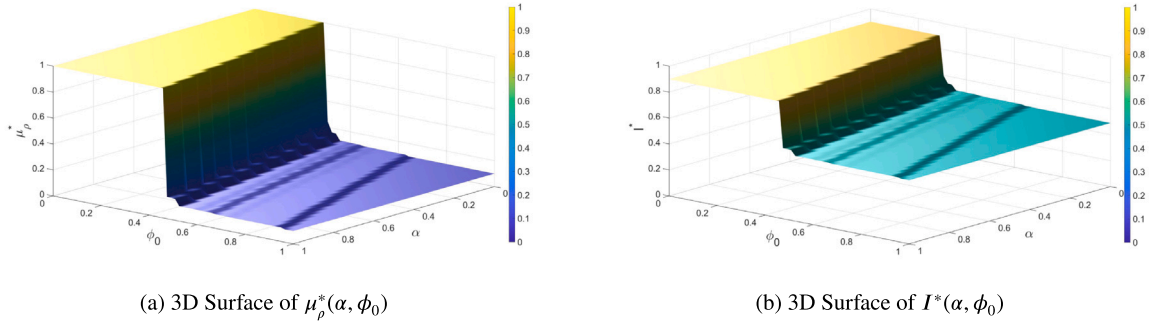


Fig. 4. 3D Surface(The X-axis is the initial opinion threshold ϕ_0 , the Y-axis is α , and the Z-axis and color both represent the final steady-state proportion of good opinions μ_ρ^* or the fraction of active consumers I^*).

Based on these observations, further investigation of the three-dimensional parameter space (ϕ_0, α, b) is carried out by fixing the parameter b , in order to examine the core regulatory role of the coupling function (5) in the opinion–consumption interaction process.

Fig. 3 employs an ER network as the first layer, where the dynamic variables ϕ_t , $\mu_{\rho,t}$, and I_t exhibit pronounced stepwise transitions. Using the same parameter set as in the simulation study, one–two iterations of the theoretically derived recurrence relations yield the fixed-point estimates $\mu^* \approx 0.1139$ and $I^* \approx 0.5933$. These estimates are consistent with the simulation outcomes $\mu^* \approx 0.1104$ and $I^* \approx 0.5933$, differing by no more than 3×10^{-3} .

Specifically, $\mu_{\rho,t}$ experiences two abrupt surges – around $t \approx 10$ and $t \approx 40$, it jumps from 0.10 to over 0.115 – while I_t steadily climbs from below 0.2 to above 0.5 and remains high after $t = 50$. Overall, these results underscore the mutual reinforcement between the opinion threshold mechanism and the consumption activation process, where under appropriate parameter conditions, individual opinions and consumption tendencies become tightly coupled and converge synchronously to a steady state.

To further elucidate the phase transition phenomena of the system over a broader parameter range, we continuously scanned the initial threshold ϕ_0 and the parameter α from 0 to 1. The results are shown in Fig. 4.

Fig. 4(a) illustrates that the steady-state positive opinion ratio, μ_ρ^* (vertical axis), exhibits a highly nonlinear “cliff-like” distribution as a function of the initial opinion threshold, ϕ_0 (front–back axis), and the consumption feedback coefficient, α (left–right axis). In regions where ϕ_0 is sufficiently low or α is sufficiently high, the top of the surface forms a plateau close to 1, indicating that nearly all individuals adopt an optimistic view of the economy. In such cases, the acceptance threshold for positive opinions is relatively low, and the suppressive effect of consumption on the opinion threshold is markedly significant, thereby facilitating a nearly universal optimistic macro state.

Conversely, a slight increase in ϕ_0 or a reduction in α prevents the opinion threshold from being effectively lowered, causing μ_ρ^* to suddenly drop from near 1 to a valley of approximately 0.1–0.2. The vertical drop – reminiscent of a “cliff” or “step” – reveals a

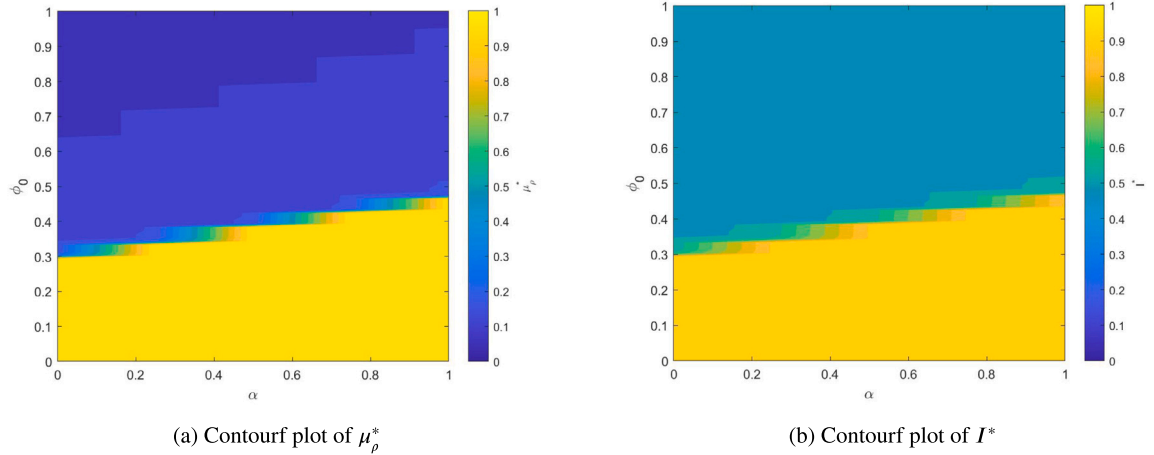


Fig. 5. Contourf plot(Horizontal axis is α , vertical axis is ϕ_0 , using shaded color bands and contour lines to indicate the values of μ_ρ^* and I^*).

characteristic phase transition. Under the positive feedback coupling of the dual-layer network, social opinion exhibits an extremely sensitive jump in response to minor parameter perturbations, suggesting that once the critical condition is exceeded, collective confidence collapses rapidly. When ϕ_0 increases further or α is very low, the entire surface remains in a deep-blue region around 0.1, indicating that the system eventually falls into a pessimistic-dominated steady state, where individuals are highly reluctant to embrace positive news and consumption fails to generate sufficient positive feedback to reverse the opinion landscape.

The abrupt transition from a plateau to a valley in this three-dimensional plot echoes real-world scenarios where a dent in optimism can precipitate a comprehensive collapse in consumption. This finding provides compelling evidence, at the macro level, for how interventions such as stimulating consumption or guiding public opinion might prevent a descent into a pessimistic trap and sustain collective confidence.

Fig. 4(b) examines the same parameter space, but with the vertical axis representing the consumption activity, I^* . Overall, I^* exhibits a bistable partition and a steep transition similar to that of μ_ρ^* . When the threshold is low (small ϕ_0) or when consumption exerts a strong influence in lowering the opinion threshold (high α), the active consumption rate can be maintained above 0.9 over time. In such circumstances, if a sufficient proportion of the population adopts an optimistic outlook on the economy, the activation probability β increases substantially, further boosting consumption participation and reinforcing optimistic opinions, thus forming a self-sustaining “consumption prosperity platform”.

However, once ϕ_0 exceeds a critical threshold or α is insufficient to counteract its influence, I^* abruptly falls from a high level to around 0.5, displaying a clear “step-like decline”. In this state, the market tends to remain cautious or adopt a wait-and-see attitude, making it difficult for spontaneous individual actions to raise consumption demand, thereby locking the system into a low-consumption regime. This observation aligns with Fig. 4(b), which demonstrates that consumption activity and optimistic opinions are closely coupled: a decline in μ_ρ leads to reduced consumption momentum, and low consumption, in turn, prevents the opinion threshold from being significantly lowered, further amplifying the spread of pessimistic views. This strong positive feedback manifests as a synchronous transition in both variables, ultimately resulting in similar high–low partitions and abrupt phase changes.

From the perspectives of economics and social dynamics, such a bistable structure implies that governments or macro-regulatory agencies could exploit the critical sensitivity near the threshold region by introducing modest interventions (e.g., encouraging consumption, tax cuts, or positive publicity) at critical moments to shift collective expectations and flip the system to a favorable high-consumption and optimistic opinion state. Otherwise, if an external shock pushes the system past the critical point, it may become trapped in a self-reinforcing “low consumption-pessimistic opinion” state.

Fig. 5(a) shows that the horizontal axis represents the consumption feedback coefficient α , the vertical axis the represents initial opinion threshold ϕ_0 , and the color indicates the steady-state proportion of positive opinions. It is evident that when ϕ_0 is low or α is sufficiently high, nearly the entire society (shown in the yellow region) holds an optimistic view of the economic outlook. In contrast, if the threshold is high and consumption feedback is lacking (blue region), most individuals tend to remain pessimistic. The sharp transition between the yellow and blue areas reveals that the system is extremely sensitive to small variations in these two key parameters. Near the critical region, even minor policy interventions or enhancements in optimistic sentiment can drive the entire population toward a highly optimistic state. Conversely, without adequate support, a widespread pessimistic collapse can occur, which explains sudden drops in macroeconomic confidence.

Fig. 5(b) employs the same parameter plane to depict the final consumption steady-state I^* . A broad expanse of warm colors indicates that, over a wide range of parameters, the overall willingness to consume remains at a medium-high to high level. However, a transition band from orange to bright yellow reveals a critical effect: when ϕ_0 is low and α is strong, individuals are sufficiently stimulated by positive opinions to quickly reach the consumption ceiling; if ϕ_0 increases or α is inadequate, the activity level can only

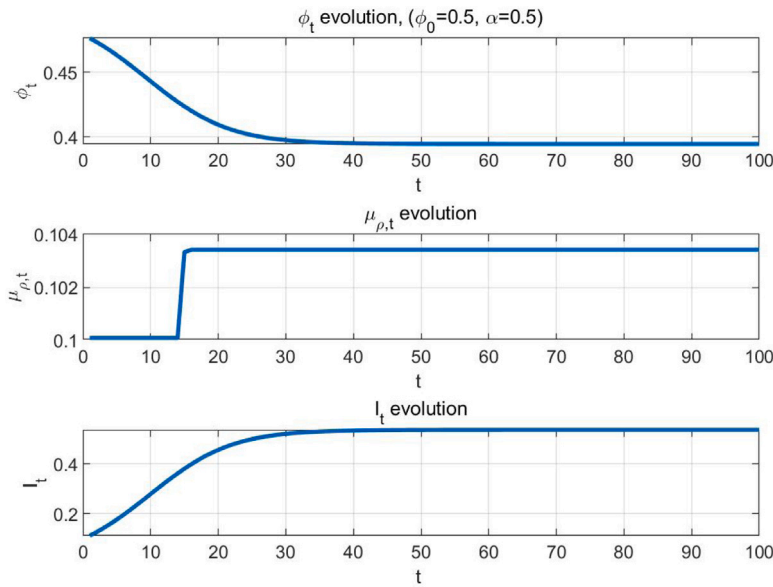


Fig. 6. Time evolution of the two-layer network: ϕ_t , $\mu_{\rho,t}$, and I_t .

be maintained at a high yet sub-maximal plateau. Together with Fig. 5(a), these results demonstrate that opinion and consumption are tightly and positively coupled: once a critical threshold is crossed, optimistic sentiment and consumption mutually reinforce each other, ultimately driving the system into a state of comprehensive prosperity; if not, society may fall into a cautious-pessimistic regime, highlighting the critical sensitivity and significant regulatory potential in policy and opinion interventions.

4.3. Model with the opinion layer as a k -regular network

When the opinion layer is configured as a k -regular network, Fig. 6 presents, from top to bottom, the evolution of ϕ_t , $\mu_{\rho,t}$, and I_t over the iteration steps. Employing the identical parameter configuration used in the numerical experiment, one-two iterations of the analytically derived mapping predict the fixed points $\mu^* \approx 0.1000$ and $I^* \approx 0.5333$. These theoretical values closely match the simulation results $\mu^* \approx 0.1034$ and $I^* \approx 0.5344$, differing by less than 3.5×10^{-3} .

In the figure, ϕ_t exhibits a marked decrease between approximately 10 and 30 steps before gradually stabilizing; $\mu_{\rho,t}$ remains low initially and undergoes a step-like jump around step 10, eventually stabilizing at about 0.104; meanwhile, I_t rapidly rises from around 0.15 to 0.45, entering a steady state together with $\mu_{\rho,t}$. Overall, these results highlight the mutual reinforcement between the opinion threshold mechanism and the consumption activation process, under appropriate parameter conditions, individual opinions and consumption tendencies become tightly coupled and converge simultaneously to a steady state.

Fig. 7(a) displays the curves of μ_ρ^* versus the initial opinion threshold ϕ_0 for different values of α (namely, 0, 0.25, 0.50, 0.75, and 1.0). It is evident that in the low ϕ_0 region, all curves remain near 1—indicating that almost all individuals are attracted by positive opinions. However, once ϕ_0 exceeds a critical value, μ_ρ^* abruptly drops to a low steady state of around 0.1, exhibiting an extremely steep transition. Notably, as α increases, the steep boundary shifts rightward, implying that stronger consumption feedback can tolerate a higher ϕ_0 while still maintaining a high proportion of optimistic views. This nonlinear jump is a typical phase transition in threshold propagation, demonstrating that the system can collapse from global optimism to widespread pessimism in response to minute parameter changes.

Fig. 7(b) depicts the evolution of the final consumption steady-state I^* as a function of ϕ_0 under the same parameter conditions. Similar to μ_ρ^* , a sharp transition emerges between a high-consumption plateau and a low-consumption valley, when ϕ_0 remains below the critical threshold, I^* reaches about 0.85–0.90, indicating a generally active consumption state; once the threshold is exceeded, I^* rapidly decays to approximately 0.5, forming a “low-consumption trap”. The different α curves also show horizontal shifts—larger α values lead to a more pronounced suppressive effect of consumption on the opinion threshold, thereby sustaining high consumption over a broader range of ϕ_0 . This further corroborates the strong positive feedback inherent in the dual-layer coupling: when society is predominantly optimistic, consumption is rapidly enhanced, and thriving consumption in turn continuously lowers the opinion threshold, further reinforcing optimism.

Fig. 8(a) presents a heatmap of the final positive opinion ratio μ_ρ^* . The color scale ranges from bright yellow (close to 1) to deep blue (around 0.1), mapping the system’s stable states as either globally optimistic or wholly pessimistic. It is evident that when $\phi_0 \leq 0.2$, the region is almost entirely yellow, indicating that even with a low α , the system easily triggers widespread propagation of positive opinions. However, as ϕ_0 increases to approximately [0.3, 0.4], the colors abruptly shift to deep blue, signifying that the system immediately falls into a low steady state with only a few individuals adopting positive opinions. Importantly, the boundary

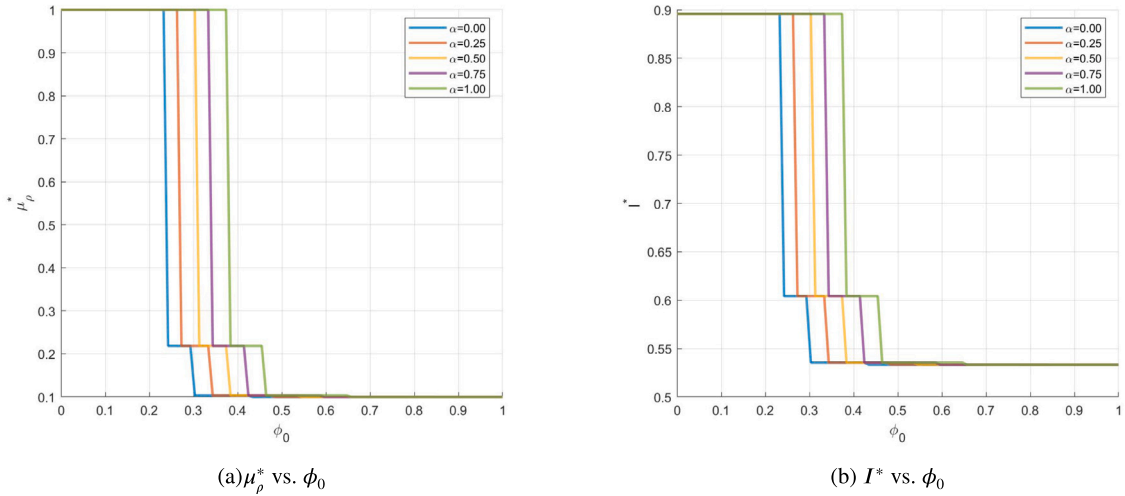


Fig. 7. Schematic of steady-state distributions (Horizontal axis is the initial opinion threshold ϕ_0 , vertical axis is the final steady-state proportion of good opinions μ_ρ^* or the fraction of active consumers I^* . Different curves correspond to different α).

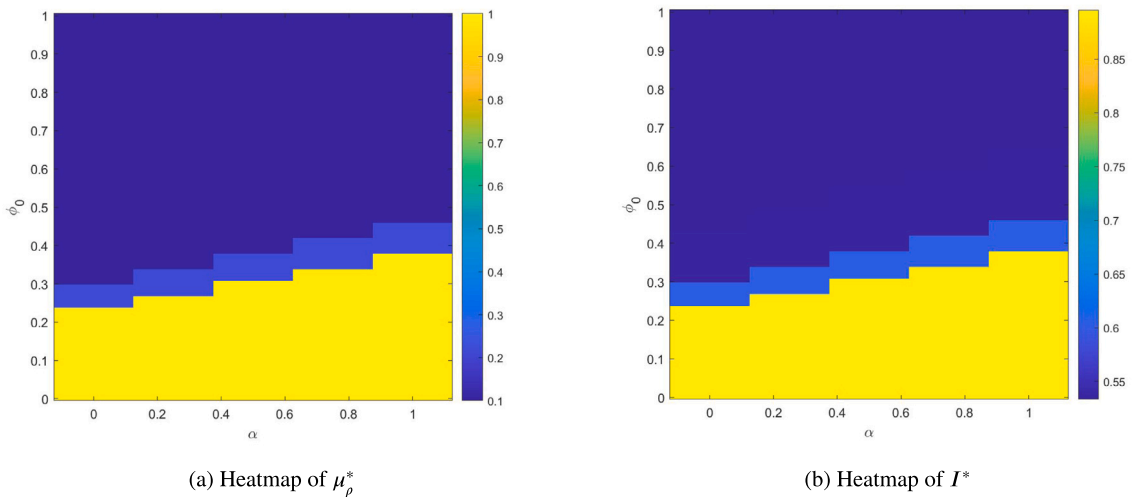


Fig. 8. Schematic of steady-state distributions (Horizontal axis is α , vertical axis is ϕ_0 , and the color indicates the final steady-state proportion of good opinions μ_ρ^* or the fraction of active consumers I^*).

between colors is nearly step-like, reflecting the discrete transition (phase change) in the system; furthermore, the boundary gradually shifts toward higher ϕ_0 as α increases, suggesting that stronger coupling allows a larger survival space for optimism.

Fig. 8(b) illustrates the distribution of the final consumption activity I^* . Its pattern is strikingly similar to that of Fig. 8(a), when ϕ_0 is low and α is high, the color is bright yellow (with $I^* \approx 0.85$ or even higher), indicating that most consumers are active; however, once ϕ_0 exceeds a certain threshold, the figure abruptly turns deep blue, meaning that consumption contracts significantly. In a macroeconomic context, this blue–yellow bistable partition echoes real-world transitions between “confidence collapse” or “depression and prosperity”, a low opinion threshold (ϕ_0) combined with sufficiently strong consumption feedback (α) can sustain a prosperous state characterized by optimism and high consumption; otherwise, widespread caution may cause the market to fall into a low-consumption trap that is difficult to reverse spontaneously.

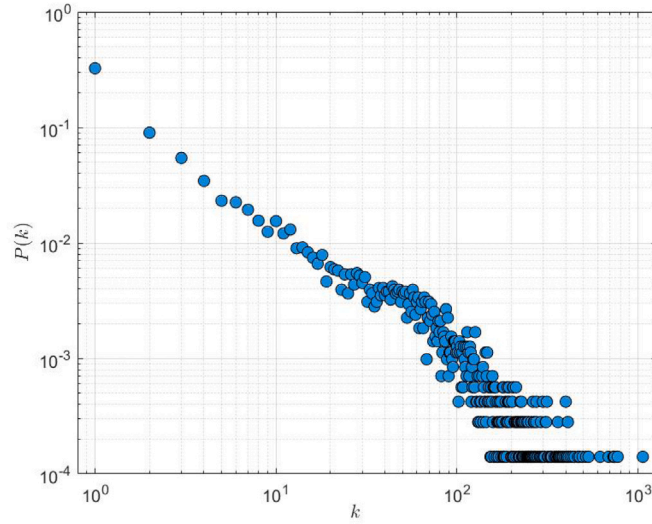


Fig. 9. Degree distribution of the Wikipedia Vote network.

4.4. Empirical validation on the Wiki-Vote network

We examine the model's external validity on the *Wiki-Vote* network <https://snap.stanford.edu/data/wiki-Vote.html>, which records the complete history of Wikipedia administrator elections from inception to January 2008 and comprises 7115 nodes and 103,689 edges. After preprocessing the data, we find that the resulting undirected graph displays a heavy-tailed degree distribution, as shown in Fig. 9. In this setting, the *Wiki-Vote* dataset is used primarily as a realistic interaction topology and does not provide direct observations of opinions or consumption activity. It is hoped that future studies will have access to more suitable datasets for modeling the co-evolution of opinions and consumption behaviors. In this real-world context, the opinion layer is directly mapped onto the empirical network, and the consumption layer is built on the same empirical graph to maintain a common vertex set. Both deterministic iterations and Monte-Carlo (MC) agent-based simulations are performed with identical parameters ($r = 0.1$, $b = 0.5$, $\beta_0 = 0.01$, $\beta_1 = 0.05$, $\gamma_0 = 0.25$, $\gamma_1 = 0.10$). The MC runs are averaged over 100 realizations to reduce sampling noise, providing a rigorous benchmark for the mean-field predictions obtained from our iterative solver. After testing the model on synthetic networks, we now evaluate it on a real-world network, and the results are shown in Fig. 10.

Panels a and b portray the analytically predicted steady states μ_ρ^* and I^* , respectively; panels c and d display their stochastic counterparts $\mu_{\rho,MC}^*$ and I_{MC}^* , all rendered as three-dimensional surfaces over the parameter plane (α, ϕ_0) , with height representing the corresponding steady-state value.

A visual inspection already reveals near-quantitative agreement. Along the high-optimism ridge ($\alpha \gtrsim 0.8$ and $\phi_0 \lesssim 0.15$) the deterministic solver yields $\mu_\rho^* = 0.91$, while the MC surface saturates at $\mu_{\rho,MC}^* = 0.87$, a relative deviation of merely $\sim 5\%$. In the diametrically opposite corner ($\alpha \simeq 1$, $\phi_0 \simeq 1$) both approaches converge to low optimism, $\mu_\rho^* = 0.13$ versus $\mu_{\rho,MC}^* = 0.10$. The consumption layer shows a comparable fidelity: at $(\alpha, \phi_0) = (1, 0.17)$ the deterministic map gives $I^* = 0.91$ and the stochastic counterpart $I_{MC}^* = 0.87$, and in the pessimistic basin their difference remains below 0.05.

The critical transition is equally robust. Tracking the sharp cliff in $\mu_\rho^*(\alpha, \phi_0)$ indicates that the tipping threshold shifts from $\phi_0 \simeq 0.17$ to $\phi_0^{MC} \simeq 0.20$, a displacement of only $\Delta\phi_0 \approx 0.03$. In other words, stochastic fluctuations inherent to the MC dynamics merely nudge the onset of the collective shift without altering its qualitative character. Taken together, these concordant surfaces provide strong evidence that the mean-field formulation captures the essential feedback loop between opinions and consumption even on a real, heterogeneously connected network, reinforcing the model's relevance for empirical socio-economic systems.

5. Discussion

5.1. Results analysis

The experimental results demonstrate high consistency across the three different network structures, including the ER- k -regular two-layer network, k -regular two-layer network, and the *Wiki-Vote* two-layer network. As the baseline sentiment level decreases, individuals become significantly more sensitive to neighborhood influences, leading to an increased probability of opinion switching. Through this mechanism, optimistic opinions rapidly spread throughout the network, further driving up the consumption layer's behavior propagation probability. This process gradually drives the system toward a high-consumption steady state, exhibiting a pronounced feedback effect and phase transition.

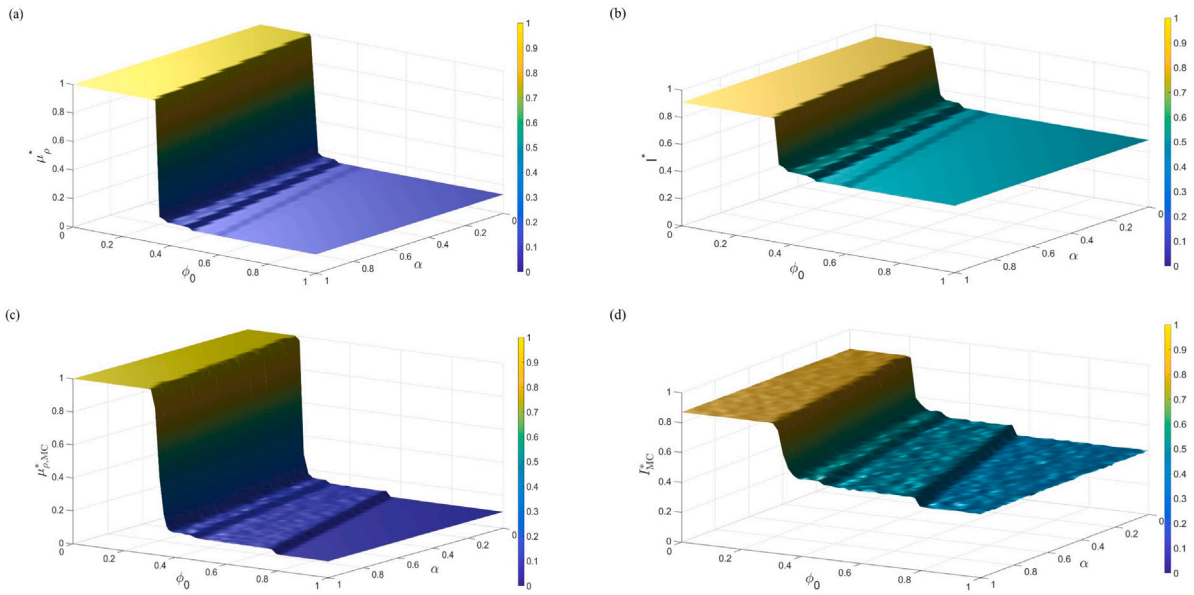


Fig. 10. Steady-state landscapes on the *Wiki-Vote* two-layer network. Panels (a) and (b) display results from the deterministic iterated map, whereas panels (c) and (d) present the corresponding surfaces obtained from 30-realization Monte-Carlo simulations.

5.2. Theoretical and practical significance

From a theoretical standpoint, the two-layer threshold-SIS architecture couples opinion diffusion with consumption activation in a single analytical scaffold, extending classical threshold and epidemic models to capture cross-layer reinforcement. The formulation clarifies how local opinion thresholds and behavioral contagion co-evolve and provides a tractable basis for exploring phase transitions in socio-economic networks.

Most earlier studies treat opinion propagation and user behavior in isolation; the present model relaxes that separation by letting consumption feedback on opinion thresholds and vice-versa. Under identical topological conditions, it exposes the parameter window in which mutual amplification emerges, thereby enriching the canonical picture of information spread with an explicit behavioral dimension.

Simulation evidence indicates that the coupled opinion–consumption feedback can precipitate system-wide realignments, whereby a modest increase in the feedback coefficient α or a slight reduction in the initial threshold ϕ_0 is sufficient to lift the network out of a pessimistic low-consumption trap and into an optimistic high-consumption state. The pronounced parameter sensitivity delineates a narrow yet actionable regulation window within which fiscal incentives or targeted information campaigns may effectively anchor macro-level confidence. At the same time, this finding furnishes a micro-level network perspective for understanding how aggregate economic sentiment forms and unravels, and it offers decision-makers a theoretically grounded rationale for timely, precision interventions – such as calibrated consumption stimuli or accelerated dissemination of positive information – to reverse adverse expectations and stabilize economic dynamics.

6. Conclusion

We propose and investigate a dual-layer network coupling model that integrates economic outlook opinions and consumption behavior within a unified dynamical framework based on via threshold mechanisms and macroscopic self-consistency. Our numerical results, encompassing both synthetic and real-world networks, confirm the presence of critical transitions and multiple steady states within the opinion–consumption system. Specifically, within certain parameter regimes, even minimal changes in baseline optimism or threshold values can trigger large-scale consumption booms, while adverse conditions may lead to a pessimistic consumption trap.

This model provides theoretical insights for macroeconomic policy and online public opinion management. When social policies or media guidance (e.g., consumption incentives) can moderately lower the threshold or enhance spontaneous optimism, the network may undergo a global transition into a “virtuous cycle” characterized by high consumption activity. These findings not only offer a micro-level perspective on the propagation mechanisms underlying macroeconomic confidence, but also furnish valuable theoretical guidance for decision-makers on how to precisely intervene in the critical region (e.g., by appropriately encouraging consumption or accelerating the dissemination of positive information) to reverse collective expectations and stabilize economic performance.

Although this study offers valuable insights, it also has several limitations that warrant further investigation in subsequent research. First, the *Wiki-Vote* dataset used here only provides a realistic network topology and does not contain direct observational

data on opinions or consumption behaviors, which imposes certain constraints on the empirical interpretation of the results. Future work could integrate the proposed mechanism with appropriate datasets, such as opinion data from surveys and transaction-level consumption records, to enable more appropriate model validation. Second, the model employs binary states and a fixed network structure. While this simplification facilitates theoretical analysis, it does not fully capture the evolutionary nature of real-world systems, where individual states and social relationships often evolve over time. Extending the framework to incorporate continuous opinion variables and time-varying higher-order network structures (e.g., hypergraphs or simplicial complexes) could better reflect real-world interactions and help examine the universality of the conclusions drawn in this study.

CRedit authorship contribution statement

Longteng Yin: Writing – original draft, Methodology, Data curation. **Juan Chen:** Methodology, Funding acquisition, Formal analysis. **Fujuan Gao:** Writing – review & editing, Formal analysis, Data curation. **Xiaoqun Wu:** Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Complete mathematical proof

A.1. Dynamic equations and basic assumptions

Let the system state variables be the triple $(I_t, \mu_{\rho,t}, R_t) \in [0, 1]^3$, whose evolution follows:

1. I_t update:

$$I_{t+1} = I_t + \beta(\mu_{\rho,t}) \langle k_2 \rangle I_t [1 - I_t] - \gamma(\mu_{\rho,t}) I_t,$$

where

$$\beta(\mu_{\rho,t}) = \beta_0 + \beta_1 \mu_{\rho,t}, \quad \gamma(\mu_{\rho,t}) = \gamma_0 - \gamma_1 \mu_{\rho,t},$$

and

$$\beta(\mu_{\rho,t}) \langle k_2 \rangle \leq 1 + \gamma(\mu_{\rho,t}), \quad \beta_0, \beta_1, \gamma_0, \gamma_1 \in [0, 1].$$

2. $\mu_{\rho,t}$ update:

$$\mu_{\rho,t+1} = r + (1 - r) \sum_{k=0}^{\infty} p(k) \sum_{c=0}^k \binom{k}{c} R_t^c (1 - R_t)^{k-c} \delta\left((1 - b) \mu_{\rho,t} + b \frac{c}{k} - \phi_t\right)$$

where $\phi_t = \frac{\phi_0}{1 + \alpha I_t}$, $\delta(\cdot)$ denotes the Heaviside step function ($\delta(x) = 1$ if $x \geq 0$, and 0 otherwise), $r, b, \phi_0, \alpha \in [0, 1]$, and

$$\sum_{k=0}^{\infty} p(k) = 1.$$

3. R_t update:

$$R_{t+1} = r + (1 - r) \sum_{k=1}^{\infty} \frac{k p(k)}{\langle k_1 \rangle} \sum_{c=0}^{k-1} \binom{k-1}{c} R_t^c (1 - R_t)^{k-1-c} \delta\left((1 - b) \mu_{\rho,t} + b \frac{c}{k} - \phi_t\right)$$

4. State space: $D = [0, 1]^3$ is a compact convex set.

A.2. Preliminary knowledge and lemmas

A.2.1. Kakutani fixed point theorem (1941)

Let $D \subset \mathbb{R}^n$ be a nonempty compact convex set, and let $\tilde{F} : D \rightrightarrows D$ be a set-valued mapping satisfying:

1. *Nonempty closed convex values:* For any $x \in D$, the set $\tilde{F}(x)$ is nonempty, closed, and convex.
2. *Upper semicontinuity:* The graph $\text{Graph}(\tilde{F}) = \{(x, y) \mid y \in \tilde{F}(x)\}$ is a closed set.

Then there exists $x^* \in D$ such that $x^* \in \tilde{F}(x^*)$.

A.2.2. Auxiliary lemma

Lemma 1 (Boundedness of the State Variables). *For any initial state $(I_0, \mu_{\rho,0}, R_0) \in \mathcal{D}$, it holds that $(I_t, \mu_{\rho,t}, R_t) \in \mathcal{D}$ for all $t \geq 0$.*

Proof.

For I_t :

From the parameter constraints $\beta(\mu_{\rho,t})\langle k_2 \rangle \leq 1 + \gamma(\mu_{\rho,t})$ and $\beta_0, \beta_1, \gamma_0, \gamma_1 \in [0, 1]$, we derive

$$I_{t+1} \leq I_t + \beta(\mu_{\rho,t})\langle k_2 \rangle I_t (1 - I_t) \leq I_t + I_t (1 - I_t) \leq 1,$$

$$I_{t+1} \geq I_t - \gamma(\mu_{\rho,t}) I_t \geq 0.$$

Hence $I_{t+1} \in [0, 1]$.

For $\mu_{\rho,t}$:

For any $k \geq 0$ and $c \in \{0, 1, \dots, k\}$, observe $\sum_{c=0}^k \binom{k}{c} R_t^c (1 - R_t)^{k-c} = 1$, so each term $\binom{k}{c} R_t^c (1 - R_t)^{k-c} \in [0, 1]$. The Heaviside function $\delta((1-b)\mu_{\rho,t} + b\frac{c}{k} - \phi_t)$ is either 0 or 1, preserving only the terms satisfying the threshold condition. Since $p(k)$ and $\binom{k}{c} R_t^c (1 - R_t)^{k-c}$ both form probability distributions, the resulting weighted sum $\sum_{k=0}^{\infty} p(k) \sum_{c=0}^k \dots \delta(\cdot)$ lies in $[0, 1]$. By linear combination:

$$\mu_{\rho,t+1} = r + (1-r) \cdot [0, 1] \subset [r, 1] \subset [0, 1].$$

For R_t :

A similar argument shows $R_{t+1} \in [r, 1] \subset [0, 1]$.

Therefore, $\mathcal{D}$ is closed under the given dynamics.

A.3. Construction of the set-valued mapping

A.3.1. Discontinuity analysis

The Heaviside function $\delta(x)$ is discontinuous at $x = 0$. This causes the mapping $F : \mathcal{D} \rightarrow \mathcal{D}$ to be discontinuous on the manifold satisfying

$$(1-b)\mu_{\rho,t} + b\frac{c}{k} = \phi_t = \frac{\phi_0}{1 + \alpha I_t}.$$

To handle this, we construct a set-valued mapping $\tilde{F} : \mathcal{D} \rightrightarrows \mathcal{D}$ defined by

$$\tilde{F}(I, \mu, R) = (I', \mu'^{\text{set}}, R'^{\text{set}}),$$

where $I' = I + \beta(\mu)\langle k_2 \rangle I (1 - I) - \gamma(\mu) I$ is a single-valued continuous function, and $\mu'^{\text{set}}, R'^{\text{set}}$ are interval-valued functions constructed as follows.

For each $k \geq 1$ and $c \in \{0, 1, \dots, k-1\}$:

- If $(1-b)\mu + b\frac{c}{k} \neq \phi$, then $\delta(\cdot)$ is fixed to 0 or 1.
- If $(1-b)\mu + b\frac{c}{k} = \phi$, we extend $\delta(\cdot)$ to the interval $[0, 1]$.

Formally,

$$\begin{aligned} \mu'^{\text{set}} &= r + (1-r) \sum_{k=0}^{\infty} p(k) \sum_{c=0}^k \binom{k}{c} R^c (1-R)^{k-c} \Delta_{k,c}(\mu, I), \\ R'^{\text{set}} &= r + (1-r) \sum_{k=1}^{\infty} \frac{k p(k)}{\langle k_1 \rangle} \sum_{c=0}^{k-1} \binom{k-1}{c} R^c (1-R)^{k-1-c} \Delta_{k,c}(\mu, I) \end{aligned}$$

where

$$\Delta_{k,c}(\mu, I) = \begin{cases} \{0\}, & \text{if } (1-b)\mu + b\frac{c}{k} < \phi, \\ [0, 1], & \text{if } (1-b)\mu + b\frac{c}{k} = \phi, \\ \{1\}, & \text{if } (1-b)\mu + b\frac{c}{k} > \phi. \end{cases}$$

A.3.2. Verification of nonempty closed convex values

Proposition 1 (Nonempty Closed Convex Values). *For any $(I, \mu, R) \in \mathcal{D}$, the set $\tilde{F}(I, \mu, R)$ is nonempty, closed, and convex.*

Proof. *Nonempty:* There is at least one selection (e.g., choosing $\delta = 0$ or $\delta = 1$ at each critical point) such that the image of $\tilde{F}$ is nonempty. *Closed:* Weighted sums of closed intervals remain closed. *Convex:* Each critical term $\Delta_{k,c}(\mu, I)$ is in $[0, 1]$, and the weighted sum of convex sets remains convex.

A.4. Rigorous proof of upper semicontinuity

Proposition 2 (Upper Semicontinuity). *The set-valued mapping $\tilde{F}$ is upper semicontinuous.*

Proof. Consider a sequence $\{(x_n, y_n)\}$ converging to (x, y) with $y_n \in \tilde{F}(x_n)$. We must show $y \in \tilde{F}(x)$. Two cases arise:

1. *Non-critical case:* If x is not on any critical manifold, there exists a neighborhood U such that $\tilde{F}|_U$ is a single-valued continuous map. Hence $y = F(x) \in \tilde{F}(x)$.
2. *Critical case:* Suppose x lies on a critical manifold. For sufficiently large n , x_n may lie in a neighborhood of x that is either critical or non-critical.
 - If x_n is eventually non-critical, then y is the limit of $F(x_n)$, and thus $y \in \tilde{F}(x)$.
 - If x_n intersects the critical manifold infinitely often, any limit point of $\{y_n\}$ is a convex combination of all possible limits, which (by the convexity of $\tilde{F}(x)$) also lies in $\tilde{F}(x)$.

Therefore, $\text{Graph}(\tilde{F})$ is closed, implying that $\tilde{F}$ is upper semicontinuous.

By Lemma 1, Proposition 1, and Proposition 2, all conditions of the Kakutani Fixed Point Theorem are satisfied. Hence there exists a fixed point $(I^*, \mu_\rho^*, R^*) \in D$ such that

$$(I^*, \mu_\rho^*, R^*) \in \tilde{F}(I^*, \mu_\rho^*, R^*). \quad \square$$

Data availability

Data will be made available on request.

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